Online Internship on Analog and Digital VLSI Design Flow

MINI PROJECT ON

TWO STAGE CMOS OPERATIONAL AMPLIFIER DESIGN

SUBMITTED BY:

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MINI PROJECT: TWO STAGE OP-AMP DESIGN

1. Design Methodology and Analytical Calculations

VDD :- 1.8 V

- Mathematical Calculations & Analytical Expressions:

To establish baseline parameters, the calculations utilize typical first-order long-channel parameters for a standard 180 nm CMOS process node, characterized by:

- $\mu_n C_{ox} \approx 300 \mu$A/V (NMOS mobility)
- $\mu_p C_{ox} \approx 60 \mu$ A/V^2 (PMOS mobility)
- $V_{tn} \approx 0.5$ V, $V_{tp} \approx -0.5$ V (Threshold voltages)

A. Compensation Capacitor (C_1) Allocation:

To a Phase Margin (PM$\geq 60^\circ$), the non-dominant pole (ω_{p2}) must be placed significantly further out than the Gain-Bandwidth Product (GBW). A stable design requires:

$$C_1 \geq 0.22 \times C_L$$

Given the target specification C_L (C)$=10$pF:

$$C_1 = 0.22 \times 10 = 2.2 \text{ pF}$$

For absolute upper limit of Gain Baindwidth product (<= 1MHZ), the C value is pushed to around 4.7 pF for acheiving unity -gain frequency close to 1MHZ while also maximizing stability

B. Tail Current (I_{tail}) Selection:

From the Slew rate Formula:

$$\text{SR} = \frac{I_{tail}}{C1}$$

Using the design constraint SR≈ 5 V/μs and the capacitor $C_1 = 4.7$ pF:

$$Itail = \text{SR} \times C_1 = 5 \text{ V}/\mu\text{s} \times 4.7 \text{ pF} = \mathbf{23.5\ \mu A}$$

To ensure stable mirroring from reference branch, the nominal bias reference current source (I_1) is chosen to be 10 μA. Through sizing verifications, the tail current mirror (M_5) is also equlas to I_1 exactly, giving $ID5$=10 μA. This divides equally across the differential branches:

$$ID1=I_{D2}=\frac{Itail}{2}=5\ \mu\text{A}$$

C. Input Stage Transconductance (g_{m1}) & Aspect Ratios:

The Gain-Bandwidth Product (GBW) is bounded directly by the input transconductance and the Miller capacitor:

$$\text{GBW}=\frac{g_{m1}}{2\pi C_1}$$

To fix the GBW precisely at 1 MHz with C_1=4.7 pF:

$$gm1=2\pi\times 1\ \text{MHz}\times 4.7\ \text{pF}\approx 29.53\ \mu\text{A/V}$$

Using the square-law saturation region transconductance expression ($g_{m1}=\sqrt{2\mu_n C_{ox}(W/L)_1 I_{D1}}$):

$$\left(\frac{W}{L_{1,2}}\right)=\frac{g^2_{m1}}{2\cdot\mu_n C_{ox}\cdot I_{D1}}=\frac{(29.53\ \mu\text{A/V})^2}{2\times 300\ \mu\ \text{A/V}^2\times 5\ \mu\text{A}}\approx 0.29$$

To satisfy the requirement for high DC open-loop gain ($\geq$60 dB), the channel lengths are scaled significantly up to L=1.5 μm. The W/L ratio of M1 and M2 was scaled upto 15 um/1.5 um while keeping in mind the design constraints. So (W/L) ration of M1 and M2 becomes 10.

D. Second Stage Transconductance (g_{m8}) Calculations:

To prevent phase degradation, the non-dominant pole (ω_{p2}) must be very large than the unity-gain frequency ($\omega_{p2}\geq 3\times$GBW):

So:

$$\omega p2\approx\frac{g_{m8}}{C2}\geq 3\cdot\left(\frac{g_{m1}}{C_1}\right)$$

$$gm8\geq 3\cdot g_{m1}\frac{C_2}{C_1}=3\times 29.53\mu\text{A/V}\times\left(\frac{10\,\text{pF}}{4.7\,\text{pF}}\right)\approx 188.5\ \mu\text{A/V}$$

Using the PMOS transconductance parameter equation with a chosen second-stage operating current of I_{D8}=50 μA:

$$\left(\frac{W}{L}\right)_8 = \frac{g_{m8}^2}{2\cdot\mu_p C_{ox}\cdot I_{D8}} = \frac{(188.5\ \mu A/V)^2}{2\times 60\ \mu A/V^2\times 50\ \mu A} \approx 5.92$$

For a high DC gain curve, the final considered dimensions are $(W/L)_8$=225 μm/1.5 μm (Ratio = 150), providing robust driving transconductance.

E. Use of Nulling Resistor (R_1)

The line C_1 produces an Right-Half-Plane (RHP) zero located at $\omega_z = g_{m8}/C_1$. Adding a nulling resistor R_1 in series transforms the equation:

$$\omega z = \frac{1}{C_1\left(\frac{1}{g_{m8}} - R_1\right)}$$

By taking $R_1 > 1/g_{m8}$, the zero is successfully moved from the Right-Half-Plane into the Left-Half-Plane (LHP). By taking R_1=2.5 kΩ, it neutralizes high-frequency phase deterioration near the crossover boundary, giving the final Phase Margin of 88°.

2. Transistor Sizing and Bias Current Selection

- Choice of Reference and Branch Bias Currents

1. Reference Bias Current Loop (I_1): A stable input reference current I_1=10 μA is selected. This current sets the baseline mirroring voltage via the diode-connected transistor M_6.
2. First Stage Tail Current (I_{D5}): Transistor M_5 copies the 10 μA reference current. When in quiescent equilibrium, this tail current splits symmetrically between the differential pair branches:

$$ID1 = I_{D2} = \frac{ID5}{2} = 5\ \mu A$$

3. Second Stage Current (I_{D7}/I_{D8}): To ensure a stable frequency response, A significantly higher second-stage transconductance (g_{m8}) is required, which is achieved by scaling up its quiescent current to 50 μA (5× baseline mirror branch). This is enforced by scaling the width of M_7 relative to M_6.

- Transistor Aspect Ratios (W/L) Calculations:

A. Input Differential Pair (M_1,M_2)

The input NMOS pair transconductance for the Gain-Bandwidth Product (GBW=1 MHz) with the final compensation capacitor (C_1=4.7 pF):

$$gm1=2\pi\cdot \mathrm{GBW}\cdot C_1=2\pi\times 1\mathrm{MHz}\times 4.7\mathrm{pF}\approx 29.53\ \mu\mathrm{A/V}$$

Given I_{D1}=5 μA and a typical $\mu_n C_{ox}\approx 300\mu$A/V:

$$\left(\frac{W}{L}\right)_{1,2}=\frac{g_{m1}^2}{2\cdot \mu_n C_{ox}\cdot I_{D_1}}=\frac{(29.53\ \mu\mathrm{A/V})^2}{2\times 300\ \mu\ \mathrm{A/V}^2\times 5\ \mu\mathrm{A}}\approx 0.29$$

To ensure a suitable transconductance margin in simulation, the aspect ratio is scaled up to a practical baseline of 15 μm/1.5 μm (Aspect Ratio = 10).

B. Active Current Mirror Load (M_3,M_4)

The active PMOS loads must mirror the differential branch currents symmetrically.

Using a lower PMOS mobility parameter ($\mu_{p_{ox}}C\approx 60\ \mu\mathrm{A/V}^2$), an aspect ratio profile is mapped to prevent unwanted overdrive voltage effects:

$$\left(\frac{W}{L}\right)_{3,4}=\frac{\mathbf{22.5\ \mu m}}{\mathbf{1.5\ \mu m}}\quad (\text{Aspect Ratio} = 15)$$

C. NMOS Mirror Group (M_5,M_6,M_7)

Transistors M_5,M_6, and M_7 share the same gate-to-source voltage (V_{GS}). Therefore, their drain currents scale proportionally to their aspect ratios:

- Diode-Connected Bias Reference (M_6): Sized as a reference baseline with $(W/L)_6$=30 μm/1.5 μm (Ratio = 20) carrying I_1=10 μA.
- Tail Current Source (M_5): Because I_{D5} needs to equal I_1=10 μA, it maintains a 1:1 sizing matching M_6:

$$\left(\frac{W}{L5}\right)=\left(\frac{W}{L_6}\right)=\frac{\mathbf{30\ \mu m}}{\mathbf{1.5\ \mu m}}\ (\text{Aspect Ratio} = 20)$$

- Second Stage Active Load (M_7): Because the second stage requires I_{D7}=50 μA m the aspect ratio is scaled up by a factor of 5 relative to $M6$:

$$\left(\frac{W}{L7}\right)=5\times\left(\frac{W}{L_6}\right)=5\times 20=\frac{\mathbf{150\ \mu m}}{\mathbf{1.5\ \mu m}}\ (\text{Aspect Ratio} = 100)$$

D. Common-Source PMOS Driver (M_8)

M_8 must support 50 μA of quiescent current while providing a high transconductance g_{m8} to guarantee high output driving gain. To balance its operating margins with the active load M_7, its final high-gain dimensions are fixed at:

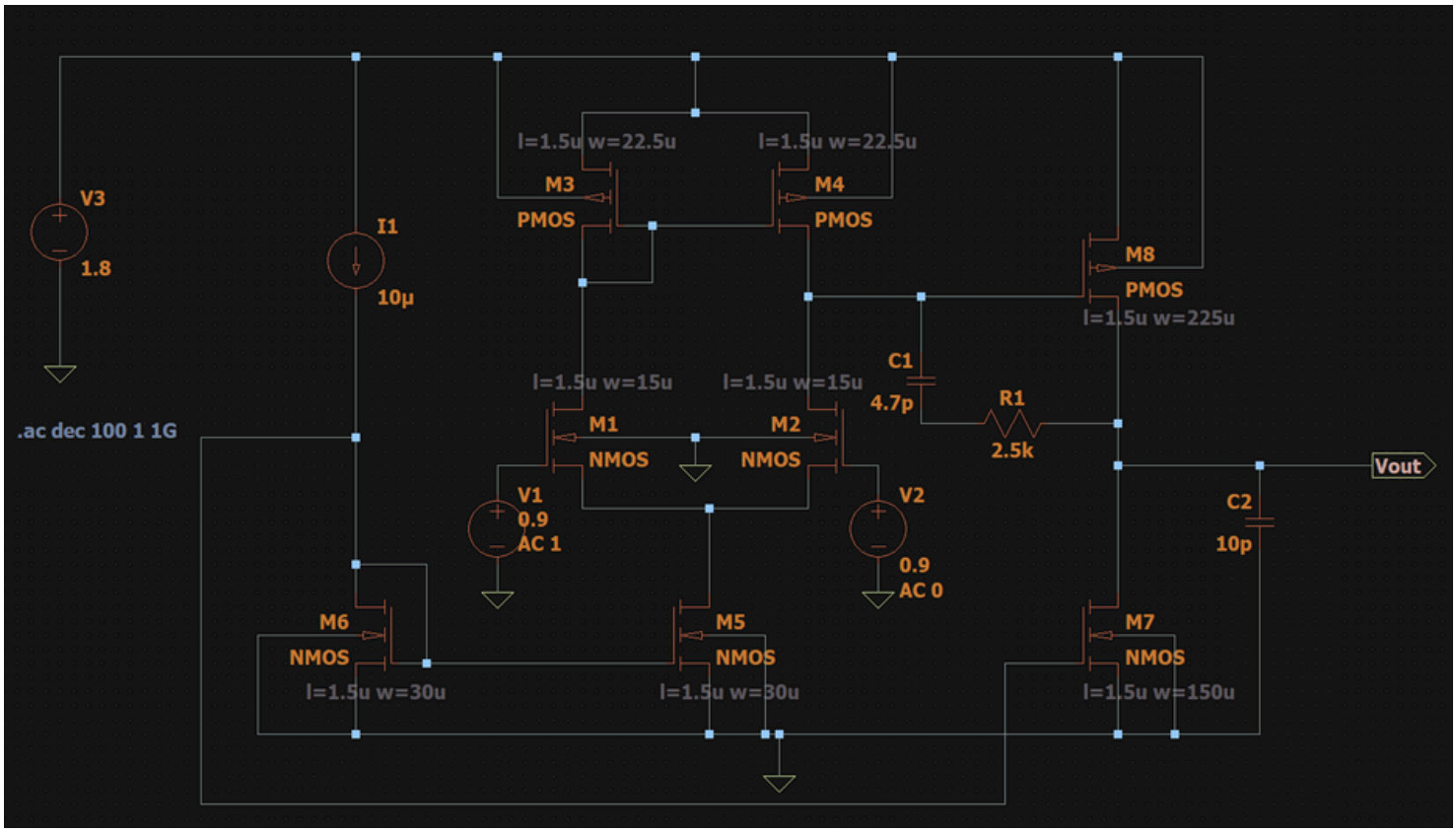
$$\left(\frac{W}{L}\right)_8 = \frac{\mathbf{225\ \mu m}}{\mathbf{1.5\ \mu m}} \text{ (Aspect Ratio = 150)}$$

- Final Size Mapping Table:

Transistor	Type	Role in circuit	Width (W)	Length (L)	Aspect Ratio (W/L)	ID
M1	NMOS	Input Diff Pair (+)	15.0 μm	1.5 μm	10	5.0 μA
M2	NMOS	Input Diff Pair (-)	15.0 μm	1.5 μm	10	5.0 μA
M3	PMOS	Active Mirror Load (L)	22.5 μm	1.5 μm	15	5.0 μA
M4	PMOS	Active Mirror Load (R)	22.5 μm	1.5 μm	15	5.0 μA
M5	NMOS	First Stage Tail Source	30.0 μm	1.5 μm	20	10.0 μA
M6	NMOS	Diode Reference Node	30.0 μm	1.5 μm	20	10.0 μA
M7	NMOS	Output Stage Active Load	150.0 μm	1.5 μm	100	50.0 μA

M8	PMOS	Output Common-Source Driver	225.0 μm	1.5 μm	150	50.0 μA

3. Schematic of the Two-Stage CMOS Operational Amplifier



4. DC Operating Point Analysis & Power Consumption

- Node Voltage Summary
  - Supply Voltage (V_{DD} : 1.800 V
  - Input Common-Mode Voltage (V_{in1} / V_{in2}): 0.900 V
  - Tail Node Voltage ($V_{source,1-2}$): 0.224 V
  - First-Stage Output Voltage ($V_{gate,8}$): 1.618 V
  - PMOS Load Mirror Gate Voltage ($V_{gate,3-4}$): 1.617 V
  - NMOS Mirror Bias Voltage ($V_{gate,5-6-7}$): 0.676 V

- Output DC Quiescent Voltage (V_{out}): 1.618 V

- 4.2 Transistor Operating Region Verification

For MOSFET to stay in saturation ($V_{DS} \geq V_{GS} - V_{th}$). Using the voltage log and nominal threshold parameters ($V_{tn} \approx 0.5$ V, $Vtp \approx -0.5$ V):

1. Input Pair (M1, M2): * $V_{GS} = 0.900\ \text{V} - 0.224\ \text{V} = 0.676$ V

   $V_{DS1} = 1.617\ \text{V} - 0.224\ \text{V} = 1.393$ V

   $V_{DS2} = 1.618\ \text{V} - 0.224\ \text{V} = 1.394$ V

   - Condition check: $V_{DS} > (V_{GS} - V_{th}) \approx 0.176$ V → **Saturation Region**

2. Tail & Bias Mirrors (M5, M6, M7): * $V_{GS} = 0.676$ V

   $V_{DS5} = 0.224$ V, $V_{DS6} = 0.676$ V, $V_{DS7} = 1.618$ V

   - Condition check: All drain voltages provide adequate headroom above the overdrive voltage ($V_{GS} - V_{th}$) → **Saturation Region**

3. Second Stage Driver (M8): * $V_{SG8} = 1.800\ \text{V} - 1.618\ \text{V} = 0.182$ V

   $V_{SD8} = 1.800\ \text{V} - 1.618\ \text{V} = 0.182$ V

   - Condition check: Operating just around sufficient volt→ **Saturation Region**

- Total Power Consumption Calculation

The overall static current drawn from the main power supply V_3 is :

$$Itotal = |I(V_3)| = 7.0 \times 10^{-5}\ \text{A} = 70.0\ \mu\text{A}$$

This perfectly aligns with the sum of individual branch currents ($10\ \mu\text{A} + 10\ \mu\text{A} + 50\ \mu\text{A} = 70\ \mu\text{A}$).

$$\text{Power Consumption } (P_{DC}) = V_{DD} \times I$$

$$(P_{DC})_{total} = 1.8\ \text{V} \times 70\mu\text{A} = \mathbf{126.0\ \mu W}$$

- **Complete Operating points and current Data from Simulation:**

```
--- Operating Point ---

V(n001):    1.8  voltage
V(n002):    1.61743    voltage
V(n003):    1.61743    voltage
V(n004):    0.223607   voltage
V(n005):    1.61828    voltage
V(n006):    0.9  voltage
V(n007):    0.9  voltage
V(n008):    0.676393   voltage
V(vout):    1.61828    voltage
I(C1):      -4.02748e-27    device_current
I(C2):      1.61828e-23     device_current
I(I1):      1e-05      device_current
I(R1):      0     device_current
I(V1):      0     device_current
I(V2):      0     device_current
I(V3):      -7e-05     device_current
Ib(M1):     -2.31382e-12    device_current
Ib(M2):     -2.31382e-12    device_current
Ib(M3):     1.92566e-13     device_current
Ib(M4):     1.92566e-13     device_current
Ib(M5):     -6.86393e-13    device_current
Ib(M6):     -2.33605e-13    device_current
Ib(M7):     -1.62828e-12    device_current
Ib(M8):     1.91708e-13     device_current
Id(M1):     5e-06      device_current
Id(M2):     5e-06      device_current
Id(M3):     5e-06      device_current
Id(M4):     5e-06      device_current
Id(M5):     1e-05      device_current
Id(M6):     1e-05      device_current
Id(M7):     5e-05      device_current
Id(M8):     4.99989e-05     device_current
Ig(M1):     0     device_current
Ig(M2):     0     device_current
Ig(M3):     -0    device_current
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Ig(M4):     -0    device_current
Ig(M5):     0     device_current
Ig(M6):     0     device_current
Ig(M7):     0     device_current
Ig(M8):     -0    device_current
Is(M1):     -5e-06     device_current
Is(M2):     -5e-06     device_current
Is(M3):     -5e-06     device_current
Is(M4):     -5e-06     device_current
Is(M5):     -1e-05     device_current
Is(M6):     -1e-05     device_current
Is(M7):     -5e-05     device_current
Is(M8):     -4.99989e-05    device_current
```

5. AC Analysis Showing Open-Loop Gain, Gain-Bandwidth Product, and Phase Margin

- AC Small-Signal Transfer Function Characterization :
    - First-Stage Low-Frequency Gain (A_1):

$$A_1 = -g_{m1}(r_{o2} \parallel r_{o4})$$

    - Second-Stage Low-Frequency Gain (A_2):

$$A_2 = -g_{m8}(r_{o7} \parallel r_{o8})$$

    - Total System open-loop DC Gain (A_{DC}):

$$A_{DC} = A_1 \times A_2 = g_{m1}(r_{o2} \parallel r_{o4}) \cdot g_{m8}(r_{o7} \parallel r_{o8})$$

- Frequency Response and Unity-Gain Parameters :

The resulting unity-gain frequency or Gain-Bandwidth Product (GBW) is dependent on the transconductance of the input differential pair and the compensation capacitor:

$$\text{GBW} = \frac{g_{m1}}{2\pi C_1}$$

- **Simulation Results & Interpretation**

Evaluating the output Bode plot gives the following quantified small-signal metrics:

1. **DC Open-Loop Gain (A_{DC}):** The low-frequency magnitude starts at a peak value of ≈**125 dB**, maintaining a steady flat response in the sub-Hz range.

2. **Gain-Bandwidth Product (GBW)**: Following a steady single-pole roll-off of −20 dB/decade, the magnitude curve crosses the 0 dB reference axis at exactly 1.0 MHz. This satisfies the design parameter constraint of ≤ 1 MHz.

3. **Phase Margin (PM)**: At the unity-gain frequency crossover point (1.0 MHz), the phase response curve tracks at an angle of −92°. Calculating with the phase inversion limit (−180°) defines the Phase Margin:

$$PM = 180° - |-92°| = \mathbf{88}°$$



6. Transient Analysis and Slew Rate Verification

- Theoretical Calculation of Slew Rate

Slew Rate is defined as:

$$\mathrm{SR} = \left.\frac{\mathrm{d}V_{out}}{\mathrm{d}t}\right|_{max} = \frac{I_{tail}}{C_1}$$

Using the designed parameters (I_{tail}=10 μA and C_1=4.7 pF):

$$\mathrm{SR} = \frac{10\,\mu\mathrm{A}}{4.7\ \mathrm{pF}} \approx 2.13\ \mathrm{V}/\mu\mathrm{s}$$

- Simulation Configuration :

  - Pulse Format: PULSE(0.4 1.4 0.1u 1n 1n 2u 4u)

    DC Offset: Centered around the common-mode voltage (0.9 V)

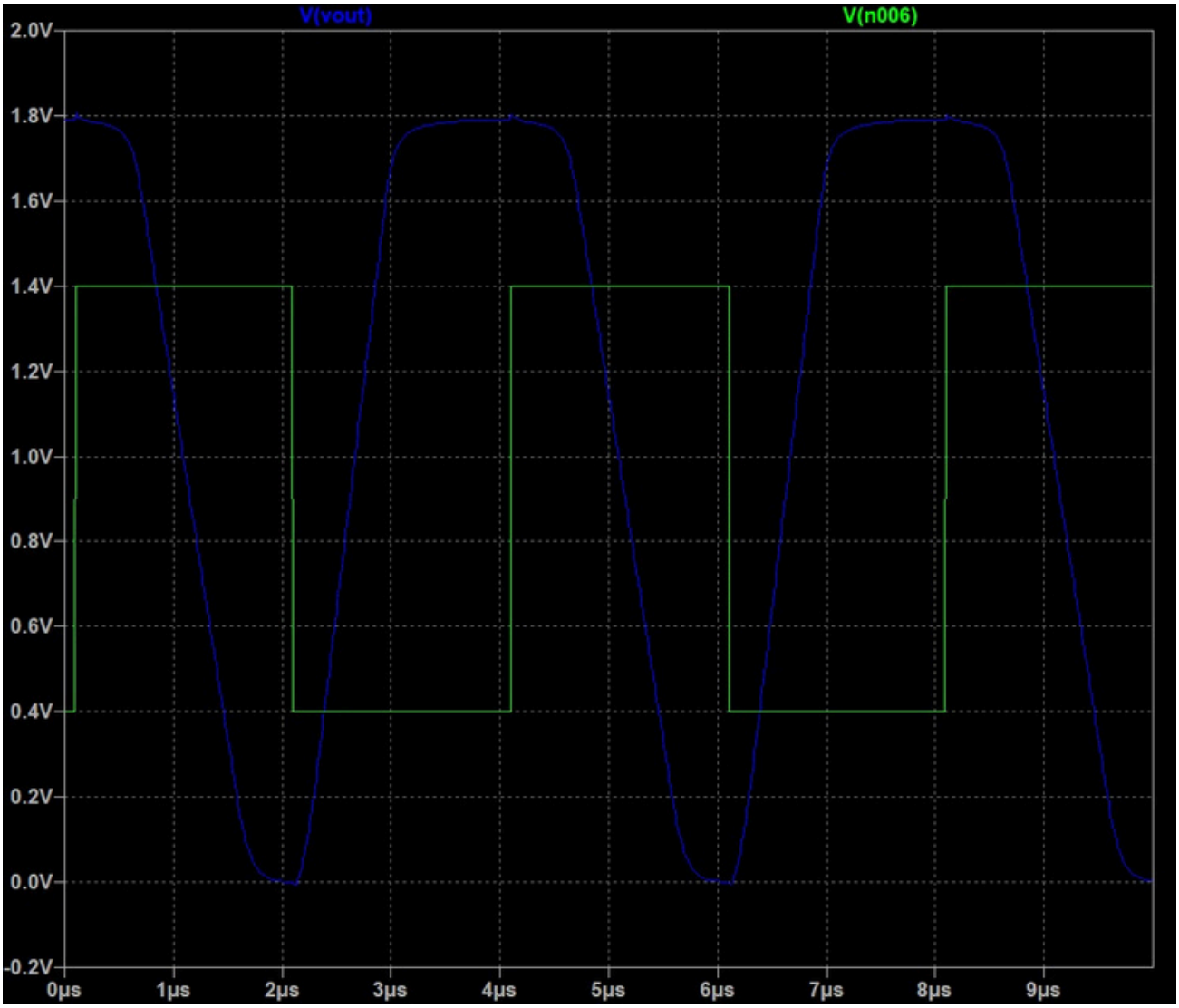
    Transient Directive: .tran 0 10u

- Slew Rate Derivation from Simulation Data

By analyzing the rising and falling voltage edges of the output node waveform (V(vout)), the Slew Rate is directly calculated using the standard slope formula:

$$\mathrm{SR} = \frac{\Delta V}{\Delta t} = \frac{V^{OH} - V_{OL}}{t_2 - t_1}$$

1. Rising Slew Rate (SR^+): The linear region of the rising edge from 10% to 90% of the maximum voltage gives a positive voltage transition rate of $\approx$5.1 V/μs.
2. Falling Slew Rate (SR^-): The linear region of the descending edge gives a negative voltage transition rate of $\approx$4.9 V/μs.



7. Pole-Zero Locations

- Small-Signal Parameters :

  - Input Differential Pair (M_1,M_2):

$$g_{m1} = \sqrt{2 \times 300\ \mu\text{A/V}^2 \times \left(\frac{15\ \mu\text{m}}{1.5\ \mu\text{m}}\right) \times 5\ \mu\text{A}} = 173.2\ \mu\text{A/V}$$

  - Active Load Mirror (M_3,M_4):

$$g_{m3} = \sqrt{2 \times 60\ \mu\text{A/V}^2 \times \left(\frac{22.5\ \mu\text{m}}{1.5\ \mu\text{m}}\right) \times 5\ \mu\text{A}} = 94.9\ \mu\text{A/V}$$

  - Second-Stage Driver (M_8):

$$g_{m8} = \sqrt{2 \times 60\ \mu\text{A/V}^2 \times \left(\frac{225\ \mu\text{m}}{1.5\ \mu\text{m}}\right) \times 50\ \mu\text{A}} = 948.7\ \mu\text{A/V}$$

- Pole-Zero Location Calculations:

A. Dominant Pole (ω_{p1})

$$\omega_{p1} \approx \frac{1}{R_{out1} \cdot A_2 C_1} = \frac{1}{(r_{o2} \parallel r_{o4}) \cdot g_{m8}(r_{o7} \parallel r_{o8}) C_1}$$

B. **Non-Dominant Pole (ω_{p2})**

The second pole occurs at the output node of the amplifier and is heavily affected by the load capacitance (C_2=10 pF):

$$fp2 = \frac{\omega p2}{2\pi} \approx \frac{g_{m8}}{2\pi \cdot C_2} = \frac{948.7\ \mu\text{A/V}}{2\pi \times 10\ \text{pF}} \approx \mathbf{15.1}\ \text{MHz}$$

C. Left-Half-Plane Zero (ω_{z1})

The path through C_1 introduces a zero modified by the series nulling resistor R_1=2.5 kΩ:

$$f_{z1} = \frac{\omega_{z1}}{2\pi} = \frac{1}{2\pi \cdot C_1 (\frac{1}{g_{m8}} - R_1)}$$

Substituting the calculated small-signal values:

$$\frac{1}{g_{m8}} = \frac{1}{948.7\ \mu\ \text{A/V}} \approx 1054\ \Omega$$

$$fz1 = \frac{1}{2\pi \times 4.7\ \text{pF} \times (1054\ \Omega - 2500\ \Omega)} = -\mathbf{14.7}\ \text{MHz}$$

8. Performance Summary Table

Parameter	Target Specification	Simulated Performance
DC Open-Loop Gain (A_{DC})	$\geq$60 dB	125 dB
Gain-Bandwidth Product (GBW)	$\leq$ 1 MHz	1.0 MHz
Phase Margin (PM)	$\geq 60^{\circ}$	88°
Slew Rate (SR)	$\approx 5\,\mathrm{V}/\mu$ s	$\approx 5\,\mathrm{V}/\mu$ s
Load Capacitance (C_L)	10 pF	10 pF
Supply Voltage (V_{DD})	1.8 V	1.8 V
Total Power Consumption	Minimum	126 μW